

MAIN SHEET Rework — Plan & Progress

Purpose of this file: this is the working plan (and live progress tracker) for the 2026-08-21 rework of Create Budget / Approve Budget / Masters against the newly-added **MAIN SHEET** in **business_knowledge/Data for R&M Portal.xlsx**. It exists so this work survives a context reset — read this before continuing the rework, and update the checkboxes as you go. Once the rework is fully verified end-to-end, fold the outcome into **CLAUDE.md** (§3/§6/§7/§9) as the single source of truth and this file can be archived/deleted.

Status as of last update: implementation code-complete and type-checks clean; NOT yet migrated/seeded against a real database, NOT yet build/runtime-tested, CLAUDE.md NOT yet updated.

Context — why this rework

The originally-built app (see **CLAUDE.md**) implemented Create Budget/Approve Budget against an earlier reading of the business rules: one **BudgetHeader** per Cost Centre + **Budget Head** + cycle, a one-level-back return chain, and a single combined AE/BE/ongoing actuals workbook. The user supplied an updated Excel workbook containing a new **MAIN SHEET** (not present when **CLAUDE.md** was last written) that changes core rules:

- **One Budget Creation Proposal per Cost Centre per FY cycle**, holding *multiple* Fund/Budget Heads and Budget Sub Heads (previously: one header per Budget Head).
- **Approval matrix** (**MAIN SHEET** rows 13–19): L1 Cost Centre User → L2 **SIC** → L3 **BIC** → L4 **TS Department** → L5 **Finance Department**. Return = "→ L1 User" at *every* level, not one-level-back. Modify: L1=Yes(Draft/Returned), SIC/BIC=No, TS/Finance=Yes.
- **New Budget Head/Sub Head list** (**MAIN SHEET** rows 26–58): a **Broad PNL Head** grouping (R&M / Power / Chemical) sits above Budget Head; 32 Budget Sub Heads across 17 Budget Heads (7 new Fund codes under Power/Chemical not in the old 25-row seed).
- New per-Budget-Sub-Head financial fields (LY Actual, Approved BE, YTD Actual — all admin-uploaded; Proposed RBE/BE — user-entered, auto-computed, (P)/(A)-tagged, fallback to Approved BE when blank) and a validation rule (RBE ≥ YTD Actual).
- A terminology mandate: **SIC, BIC, Base/LOC, Budget Sub Head** used consistently everywhere; no loose "Description"/"Category" labels.
- Admin-controlled FY: exactly one open **BudgetCycle** drives "CFY"/"NFY" labeling app-wide.

Decisions confirmed with the user (via AskUserQuestion, this session)

1. Every return (SIC/BIC/TS/Finance) goes all the way back to L1/Draft — replacing one-level-back entirely.
2. Existing **BudgetHeader/BudgetEntry/ApprovalAction/BudgetAttachment** data is **wiped** as part of the schema change (masters, Employees, Users/logins untouched).
3. The single combined actuals workbook is replaced by **three separate upload flows** (LY Actual, Approved BE, YTD Actual), each with its own downloadable sample template.
4. Both Proposed RBE and Proposed BE fall back to the Approved BE figure, literally, when the Location User leaves them blank.
5. A Budget Sub Head may appear as **multiple rows** within one proposal (not unique) — aggregation for the fallback/(P)/(A)/validation logic happens per Sub Head across all its rows.

6. A bulk Excel upload for assigning Location Users to Cost Centres was built alongside the existing one-at-a-time Authorization screen.

Progress checklist

1. Schema rewrite — done

`webapp/prisma/schema.prisma`:

- `BroadPnlHead` enum (`RM` | `POWER` | `CHEMICAL`) added; `BudgetHead.broadPnlHead` field added.
- `BudgetLineItem` renamed to `BudgetSubHead` (`code` = Fund code, `name` = Budget Sub Head text, dropped the old `description` naming).
- `BudgetHeader` **no longer has `budgetHeadId`** — unique constraint is now `@@unique([costCentreId, cycleId])` (the MAIN RULE).
- `BudgetEntry.lineItemId` → `subHeadId` (FK → `BudgetSubHead`); `itemDescription` field dropped. No uniqueness on `[headerId, subHeadId]` — multiple rows per Sub Head are allowed (confirmed).
- `ActualsDataType` renamed: `ACTUAL_EXPENDITURE` → `LY_ACTUAL`, `ONGOING_EXPENDITURE` → `YTD_ACTUAL`, `APPROVED_BE` unchanged. `ActualsImportBatch` gained a `dataType` field. `ActualsRow` simplified to `companyCode`, `costCentreCode`, `costCentreName`, `subHeadCode`, `subHeadName`, `fiscalYear`, `amount`, `dataType` (dropped unused columns: `glAccount`, `commitmentItem`, `commitmentItemName`, `fundsCenterName`, `lineItemName`, `valTypeText`, `amountType`, `period` — none were consumed downstream).

 **NOT yet done: the actual migration has not been run against a database.** Next step: `npx prisma migrate dev --name main_sheet_rework` (or, since the local `.pgdata` Postgres instance needs to be started first per `CLAUDE.md` §8 — check `pg_ctl ... status` before assuming it's running). Since existing transactional data is being wiped by agreement, it's fine to let the migration drop/recreate the transactional tables rather than hand-write a data-preserving migration.

2. Seed rewrite — code done, not re-run

`webapp/prisma/seed.ts` now parses `MAIN SHEET` rows 26–58 (columns: SN, Broad PNL Head, FUND, Budget Head, Budget Sub Head) instead of the old `Fund Centre` sheet. Asserts 34 Cost Centres / **17 Budget Heads / 32 Budget Sub Heads** / 311 Employees. `Location Mapping` and `Data base Employee` sections unchanged (verified by inspection that `MAIN SHEET` itself has no location data — its used range stops at row 58/column F, and `Location Mapping`'s "Location User" column header has no data under it — so `Location Mapping` remains the authority for Cost Centres, **correcting** a literal reading of the original request that said location mapping should come from `MAIN SHEET`). **Needs to be run** (`npx prisma db seed`) after the migration above.

3. Shared libs — done

- `src/lib/labels.ts` (**new**) — `ROLE_LABELS` (SIC/BIC/etc.), `ROLE_FULL_NAMES`, `BROAD_PNL_HEAD_LABELS`, `BASE_LOC_LABEL`, `BUDGET_SUB_HEAD_LABEL`, `formatCycleLabel/cfyLabel/nfyLabel`.
- `src/lib/cycle.ts` (**new**) — `getActiveCycle()` (the one open cycle), `setActiveCycle()` (enforces at most one open cycle at a time — auto-closes others), `previousFinancialYear()` (FY string arithmetic for LY Actual lookups).

- `src/lib/workflow.ts` — `returnOneLevel` replaced with `returnToL1`; added `MODIFY_ROLE_FOR_STATUS` and `canEditHeader(access, header)` (the one shared editability check used by both Create Budget and Approve Budget: `DRAFT+LOCATION_USER`, or `PENDING_TS+TS_DEPT`, or `PENDING_FINANCE+FINANCE_DEPT`).
- `src/lib/actuals.ts` (**new**, replaces the deleted `actuals-import.ts`) — `parseLyActualWorkbook/parseApprovedBeWorkbook/parseYtdActualWorkbook` (share one row parser, differ only in `dataType`), `buildActualsSampleWorkbook()` for the downloadable templates. Column format (7 columns: Company Code, Cost Centre Code, Cost Centre Name, Budget Sub Head Code, Budget Sub Head Name, Fiscal Year, Amount) is my own call per the user's "decide the format" instruction — trimmed to exactly what's consumed downstream.
- `src/lib/sub-head-actuals.ts` (**new**) — `getSubHeadActualsMap(costCentreCode, cycle)`, the shared LY Actual/Approved BE/YTD Actual lookup used by both the Create Budget entry grid and its save-time validation.
- `src/lib/reports.ts` — updated for the renamed enum values and switched from an ad hoc "most-recently-created cycle" default to `getActiveCycle()`.
- `src/lib/attachments.ts` — `MAX_ATTACHMENT_SIZE_BYTES` raised to 30MB, `ALLOWED_ATTACHMENT_EXTENSIONS` whitelist added (`.pdf .xls .xlsx .doc .docx .ppt .pptx .csv`), `isAllowedAttachmentName()` helper.

4. Admin — Masters — done

- **Funds tab:** `BudgetHead` create form gained a `broadPnlHead` select; listing now grouped by Broad PNL Head. `BudgetSubHead` create/delete (renamed from `BudgetLineItem`).
- **Actuals tab:** three separate upload sections (LY Actual / Approved BE / YTD Actual), each with its own "Download sample template" link and its own history rows filtered by type.
- **Settings tab:** single-open-cycle enforcement wired in (`setActiveCycle`); labels switched to CFY/NFY; new cycles are created closed by default, admin opens explicitly.
- **Roles tab:** replaced the old static 5-row table with the MAIN SHEET L1–L5 matrix verbatim (SIC/BIC labels, Modify/Return/Approve columns).
- **New Employees tab:** full CRUD (create/edit/delete) over the `Employee` master — this is what "admin can add/edit/update the Employee master" maps to.
- **Locations tab:** "Base" column header → "Base/LOC".

5. Admin — Authorization — done

- `AddAccessGrantForm` LOCATION scope is now a multi-select (creates one `UserAccess` row per selected Cost Centre in one submission) — no schema change needed.
- **New:** `BulkAssignForm` + `bulkAssignAccess` Server Action — upload an Employee No / Role / Cost Centre Code sheet (template at `/api/templates/user-location-mapping`) to add many access grants at once. Only adds grants to employees who **already have a login** — does not create logins itself (reports per-row errors for employees without one).

6. New API template routes — done

`/api/templates/ly-actual`, `/api/templates/approved-be`, `/api/templates/ytd-actual`, `/api/templates/user-location-mapping` — all admin-gated GET routes generating a downloadable `.xlsx` on the fly via `exceljs`.

7. Create Budget — done

- `create/page.tsx` + `create/actions.ts`: picker simplified to **Cost Centre only** (FY is admin-controlled via the one open cycle; Budget Head selection now happens inside the proposal page). `openBudgetHeader` finds-or-creates by `[costCentreId, cycleId]`.
- `[id]/page.tsx`: header block is fully read-only for everyone; editability of the entry grid now comes from `canEditHeader()` (so TS/Finance can edit in-place during their pending stage, not just the original Location User in Draft). Added a "History" section (shared component, see below).
- `[id]/EntryGrid.tsx` (**major rewrite**): "Add Budget Head" control (grouped by Broad PNL Head) → per-head card → **Sub Head summary table** (LY Actual/Approved BE/YTD Actual read-only + Proposed RBE/BE aggregated across that Sub Head's rows, (P)/(A)-tagged, falling back to Approved BE when blank, red-highlighted when below YTD Actual) → the entry-row editor below it (" + Add Row" picks a Budget Sub Head, repeats allowed). Attachments now 30MB / whitelisted types.
- `[id]/actions.ts`: `saveDraftEntries` reworked for `subHeadId`; added RBE ≥ -YTD-Actual validation (aggregated per Sub Head, only when the entered sum is nonzero) and a negative-amount guard; added **field-level audit diffs** (`MODIFY_ENTRY/ADD_ENTRY/DELETE_ENTRY` on `AuditLog`, entityType `BudgetHeader`) whenever the actor is on the TS/Finance modify path (not the original Draft author); `submitBudget` now explicitly rejects being called from any status other than `DRAFT`.

8. Approve Budget — done

- `actions.ts`: `actOnHeader` now calls `returnToL1` — any return lands back on `DRAFT` at L1, `currentLevel` resets to 0.
- `page.tsx`: when the viewer can `canEditHeader()` (i.e. holds `TS_DEPT/FINANCE_DEPT` for a header currently pending at their level), renders the **same** `EntryGrid` component used by Create Budget (editable) instead of the old read-only list — SIC/BIC still get the read-only list + remarks form. Added the shared History section.

9. Shared History component — done

`src/components/History.tsx` (**new**) — merges `ApprovalAction` (approve/return timeline) with the new `MODIFY_ENTRY` audit rows into one timeline; used on both `/budgets/[id]` and `/approvals` so "who returned when with what comment" and "who changed which field" are both visible in one place, on either screen.

10. Home Dashboard & Reports — done

Reworked for the schema change (Budget-Head-level grouping now derives via `entry.subHead.budgetHead` since `BudgetHeader` no longer carries `budgetHeadId` directly); `ActualsDataType` references renamed; both switched from ad hoc "most recent cycle" logic to `getActiveCycle()`; FY labels switched to `formatCycleLabel/cfyLabel/nfyLabel`; Home's KPI tiles relabeled LY Actual / Approved BE (CFY) / Proposed RBE (CFY) / YTD Actual (CFY) / Proposed BE (Nfy).

11. Terminology pass — mostly done, some gaps likely remain

Swept: nav/labels module, Locations tab, Home dashboard filters, Roles tab, Topbar role summary (was showing raw enum values like `STATION_INCHARGE` — fixed to use `ROLE_LABELS`). **Not exhaustively re-checked:** `CreateUserForm.tsx` (Authorization), `ApprovalActionForm.tsx`, and any other small components not directly touched during the rewrite — worth a final grep pass for `STATION_INCHARGE/BASE_INCHARGE` rendered raw, or "Base" used unqualified, before calling this fully done.

12. CLAUDE.md update — not started

CLAUDE.md's §3 (decisions), §6 (data model), §7 (routes), §9 (build order/status) still describe the **pre-rework** schema/workflow. Needs a full pass once the rework is verified end-to-end, per the project's own convention of keeping CLAUDE.md as the single source of truth. Do this last, after verification below — don't document behavior that hasn't actually been run yet.

What's verified so far vs. not

Verified:

- `npx prisma generate` — clean, against the new schema.
- `npx tsc --noEmit` — clean (after fixing one `AuditLog.diff` Json-typing issue by serializing via `JSON.parse(JSON.stringify(...))`, matching the pattern already used elsewhere in the codebase).

NOT yet verified — this is the actual remaining work, roughly in order:

1. Start the local Postgres instance (`.pgdata`, port 5433 — see `CLAUDE.md` §8; check `pg_ctl status` before assuming it's up or down).
2. Run `npx prisma migrate dev --name main_sheet_rework` — this is a breaking migration (drops `budgetHeadId` off `BudgetHeader`, renames tables/enums) and **will wipe existing transactional data**, which was explicitly agreed with the user. Watch for it prompting about data loss and confirm.
3. Run `npx prisma db seed` and confirm the printed counts match 34 / 17 / 32 / 311.
4. Re-run (or run for the first time on this schema) `npm run db:bootstrap-admin` if the admin user doesn't already exist post-migration — check first, don't assume.
5. `npm run build` — the real end-to-end Next.js build (lint + type-check + prerender), not just `tsc --noEmit`.
6. Manual/DB-level walkthrough: open one Cost Centre's proposal, add entries under 2+ Budget Heads, submit, SIC approve, BIC approve, TS approve **and edit an entry's RBE in place** (confirm a `MODIFY_ENTRY` audit row is written and visible in History), Finance approve, confirm `APPROVED`. Separately: confirm a return from any level lands back on `DRAFT/L1`, not one level back.
7. Confirm the RBE-≥-YTD-Actual validation actually blocks/allows saves correctly, and that the `(P)/(A)/fallback-to-Approved-BE` display logic is correct in all three states (blank, submitted, approved).
8. Upload one file through each of the three new actuals endpoints (LY Actual / Approved BE / YTD Actual) and confirm the downloaded sample template's columns match what each parser accepts.
9. Confirm `getActiveCycle()/setActiveCycle()` really does enforce exactly one open cycle, and that Home/Reports/Create Budget all agree on it.
10. Try the bulk Authorization upload (`/api/templates/user-location-mapping`) end-to-end against a real employee that already has a login.
11. Finish the terminology sweep noted in §11 above.
12. Update `CLAUDE.md` once all of the above is actually confirmed working — not before.

Nothing has been committed to git for this rework yet — check `git status/git diff` before assuming otherwise; this rework is on top of whatever branch state existed when it started (`build/initial-implementation` per the session's git context).